



DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: DATA SCIENCE – (24AD3107)

A.Y. 2026 - 2027

TITLE OF THE PROJECT		Health Scope" – A Full-Lifecycle Public Health Analytics & Risk Dashboard	
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Abstract

Health Scope is a full-lifecycle public health analytics and risk prediction system designed to analyse healthcare and demographic data and generate meaningful, data-driven insights. The project aims to address the challenges involved in processing and understanding large healthcare datasets by providing an integrated pipeline for data collection, preprocessing, analysis, machine learning, and visualization.

The system will collect health-related data from publicly available datasets such as the National Family Health Survey (NFHS), hospital admission datasets, or suitable disease-specific datasets. The collected data will undergo data ingestion, cleaning, missing-value handling, duplicate detection, outlier analysis, exploratory data analysis, and feature engineering. Machine learning algorithms including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbours, and Naïve Bayes will be trained and compared to identify the most suitable model for health-risk prediction.

The proposed system will provide an interactive dashboard for exploring health trends, demographic patterns, risk factors, and model performance. Users will be able to analyse important health indicators and obtain risk predictions based on the selected input parameters. Model performance will be evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrix.

Health Scope aims to demonstrate how data engineering, data analytics, machine learning, and interactive visualization can be integrated into a single end-to-end platform for public health analysis. The system is expected to help identify significant health patterns and risk factors and present them in an accessible form to support data-driven analysis and decision-making.